Sequential Probabilistic Inference of US 10-Year Treasury Yield Regimes for Conditional ASX Energy Stock Analysis

COMP3018 – Probabilistic Machine Learning

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Artefact 2

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Capstone and Team Integration

The overarching system that is being built is a dashboard to assist a Quantitative Trader in decision making over energy stocks. Each individual will a Hidden Markov Model that will be predicting daily change in Australian Energy Stocks via latent states. This report will be analysing the affect that the US 10-Year Treasury Yield has on Woodside Energy Stock at open of trade. The dashboard created will have daily predictions for 9 different indicators and is designed to give a Quantitative Trader uncertainty information and not just base point recommendation. The Trader will be able to see the individual Markov models and the level of risk associated with how certain the change is predicted to be.

Introduction

Problem Statement

Artefact 1 established that the US 10-Year Treasury yield shows non-stationary behaviour, with heavy tailed daily changes and time varying volatility. The Bayesian and Frequentist estimation produced a single posterior distribution over parameters, treating the entire seven-year sample as one stationary population. While this captured the overall distribution shape of yield changes it couldn't distinguish a difference from a calm day in 2019 to a volatile day during the COVID crisis. The EDA in Artefact 1 identified 3 distinct macroeconomic regimes; stable, transitional and hiking/crisis that this single population approach simply cannot represent. The objective for Artefact2 is to explicitly model the latent regime structure driving these changing parameters over time and test whether that regime has any conditional relationship with Woodside Energy Stock movements.

Sequential System Rationale

A single stationary distribution cannot account for this behaviour, Artefact1 showed clear clustering in the rolling 21day volatility, especially in areas of COVID spike and the hiking-cycle plateau. This volatility clustering is heavily documented in statistical financial research, first documented in financial price data by Mandelbrot in 1963[1].

Further evidence against a single stationary Normal distribution is found in the kurtosis and Student-t fit evaluated in Artefact1. The kurtosis of 2.14 confirms heavy tails are present in the US 10-Year Treasury Yield, and the degrees of freedom found in the Bayesian and Frequentist methods were 5.52 and 5.27 respectively. With degrees of freedom sitting around the 5 mark this is evidence that the data will see extreme events happen far more often than a normal distribution would predict. Critically, this estimate was fit across the entire seven-year window, implicitly assuming tail risk was constant over time, an assumption contradicted by the shifting volatility evident across the three macroeconomic regimes.

The limitations described above motivate the use of a sequential probabilistic model. Markov models address this directly by allowing the statistical properties of the series to shift depending on which underlying state the process currently occupies, rather than assuming a single fixed distribution across the sample time. Hamilton (1989) formalised this framework, showing that the parameters of time series models could be governed by a discrete-state Markov chain were transitions between states follow fixed probabilities[2]. This finding has since been validated, Ang and Bekaert (2002) who demonstrated that regime-switching models outperformed single-regime models in forecasting US-short rate behaviour and that regimes could correspond to business cycles[3], which directly relates to the macroeconomic regimes identified in the US 10-Year Treasury Yield.

A Hidden Markov Model is appropriate for this study's circumstances as the regime at any given time is not directly observable. In our data contract from Artefact1, there is no label indicating which regime style the trading day is in. These states must be inferred from the observed changes in yield and rolling volatility. The Hidden Markov Model separates the model into two layers; a hidden state sequence governed by Markov transition probabilities, and an observed emission sequence, whose distribution is conditioned on the current hidden state. Parameters are estimated via the Baum-Welch algorithm without requiring any labelled regime information, making this an unsupervised approach.

Using a Bayesian Information Criterion (BIC) comparison between K=2 and K=3 states, K=3 was selected. K=3 had 24 free parameters compared to 14 in K=2 and achieved a BIC of -16429 compared to -15270 for K=2, confirming the three-regime structure independently identified in the Artefact 1 exploratory data analysis.

Dataset and Modelling

Modelling

For the data pre-processing, the exported csv from Artefact1 was used. This csv contains three features indexed by date; yield_pct, daily_change and vol_21d. As the first 20 days contain Nan values for vol_21d, these rows were dropped from our dataset. Following this procedure the ASX data for Woodside energy was fetched using yfinance. The close and open prices were fetched into a dataframe. For this study we want to see how the performance of the US 10-Year Treasury yield could affect Woodside energy stock, therefore the two dataframes containing the data are inner joined on dates both marked that have been traded. Like in Artefact1 to account for the difference between Australian and US public holidays data has been forward filled. The next change made is to align the next-day Woodside opening return by shifting it back one day such that each row pairs today's yield observation with the following days opening return. The US bond market closes several hours before the ASX opens, meaning overnight yield movements are known information at time of the next ASX open.

State Space

This model uses a 3-latent state space. These state spaces directly correspond to our 3 regimes identified in Artefact1 (2018-19 moderate/low-vol, 2020-21 near zero COVID supressed and 2022-2024 hiking cycle). The state labels were assigned to each state (0: stable, 1: transitional ,2:hiking/crisis) by sorting the three learned states in ascending order of their vol_21d emission variance. The Viterbi-decoded regime across 1705 trading days was;

State 0: Stable – 602 days
State 1: Transitional – 586 days
State 2: Hiking/Crisis – 517 days

This broad and even range of days in each state demonstrates that one particular state/regime did not dominate the sample, this rules out the possibility of the model generalising to one particular dominant state in a "catch-all" way. This is consistent with the findings in Artefact1.

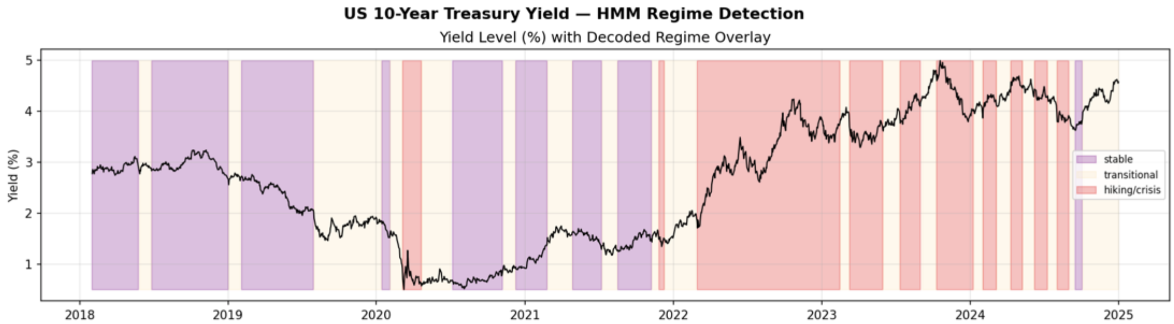
Transition Probabilities

The Transition Matrix derived from the model fitting using Baum-Welch;

	Stable	Transitional	Hiking/Crisis
Stable	0.984182	0.015818	0.000000
Transitional	0.014681	0.967149	0.018169
Hiking/Crisis	0.000000	0.020034	0.979966

As can be observed from the transition matrix, once a regime has started there is a very high chance that it will stay in this state. This regime persistence is evidence of sequential dependency in the yield series, a property that a single stationary distribution cannot capture. By looking at this matrix, it can be observed that transitions from stable to hiking/crisis or vice versa has a probability of 0%, the market never jumps from a stable regime to a hiking/crisis regime or vice versa . This means that every transition has to pass through the transitional regime. This is confirmed by the plot;

Figure 1



Using 1/(1-p) to calculate the expected regime duration;

Stable: approx. 63 trading days
Transitional: approx. 30 trading days
Hiking/Crisis: approx. 50 trading days

Emission Probabilities

The following table for Emission Parameters by State

	Mean daily change	Mean vol_21d	Vol_21d variance
Stable	-0.000479	0.033076	0.000044
Transitional	0.001724	0.050787	0.000050
Hiking/Crisis	0.000055	0.080408	0.000326

By observing the emission parameters, it can be seen that the stable regime has the lowest mean rolling volatility, this makes sense considering that during this regime it can be expected that there is little to no change. The vol_21d variance is also extremely low, showing that in the stable state changes in the daily yield are minor and consistent.

In the Transitional state, the highest mean daily change is observed, meaning that on average the largest average daily increases occurred during the transitional phases. This makes sense due to the transitional phases representing when the market was adapting to shift direction.

An interesting observation in the table is that the mean daily change in the hiking crisis is the smallest, however the variance in vol_21d is the greatest, almost 7 times greater than the other two states. This starts to make sense when we consider that the hiking/crisis regime captures big changes in either direction. With the Mean daily change being close to 0, it can be suggested that the difference of positive and negative change in this state is almost even. This confirms that regime separation in this model is driven primarily by the volatility rather than the direction of yield movement.

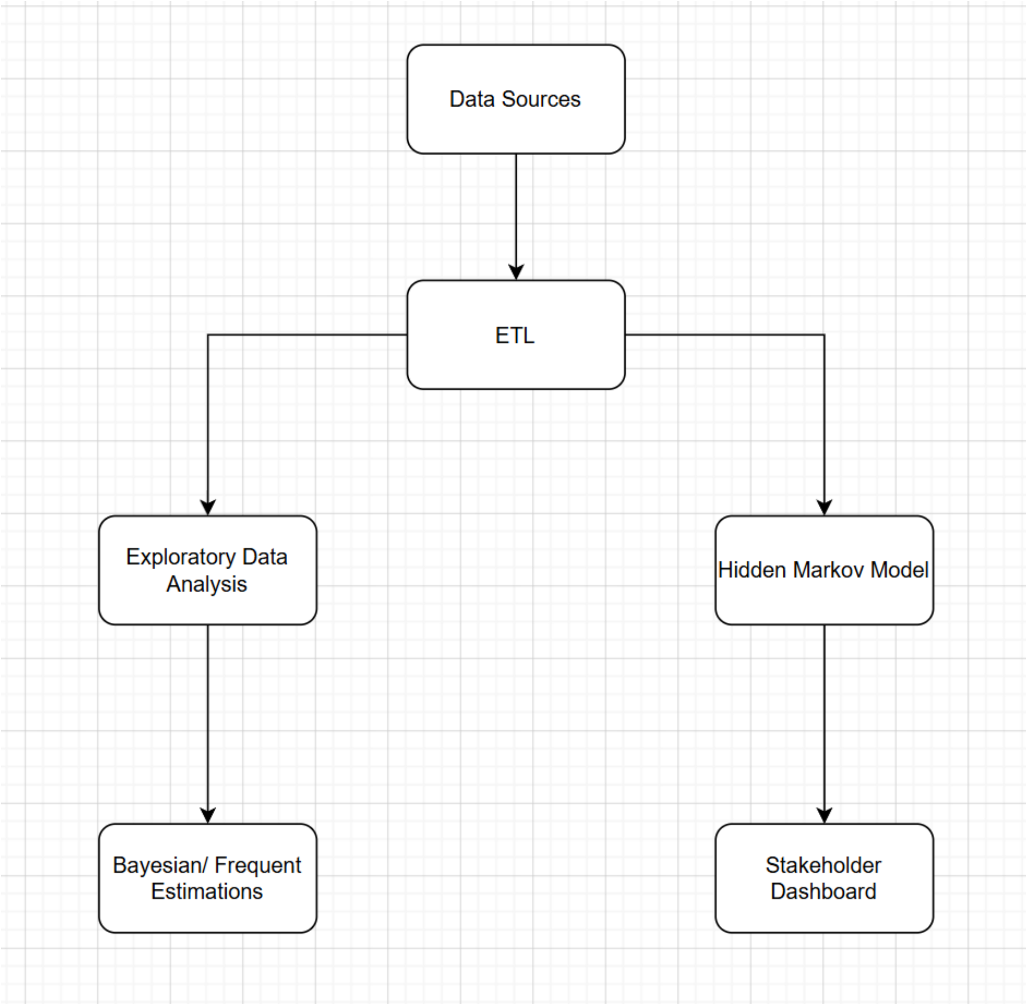
It is important to note that vol_21d is computed as a 21-day rolling standard deviation of daily_change. Meaning that the two emission features are not independent. This technically violates the assumption of conditionally independent variables. This was an accepted trade-off due to the including of volatility improved regime separation.

Why Standard Approaches are Insufficient

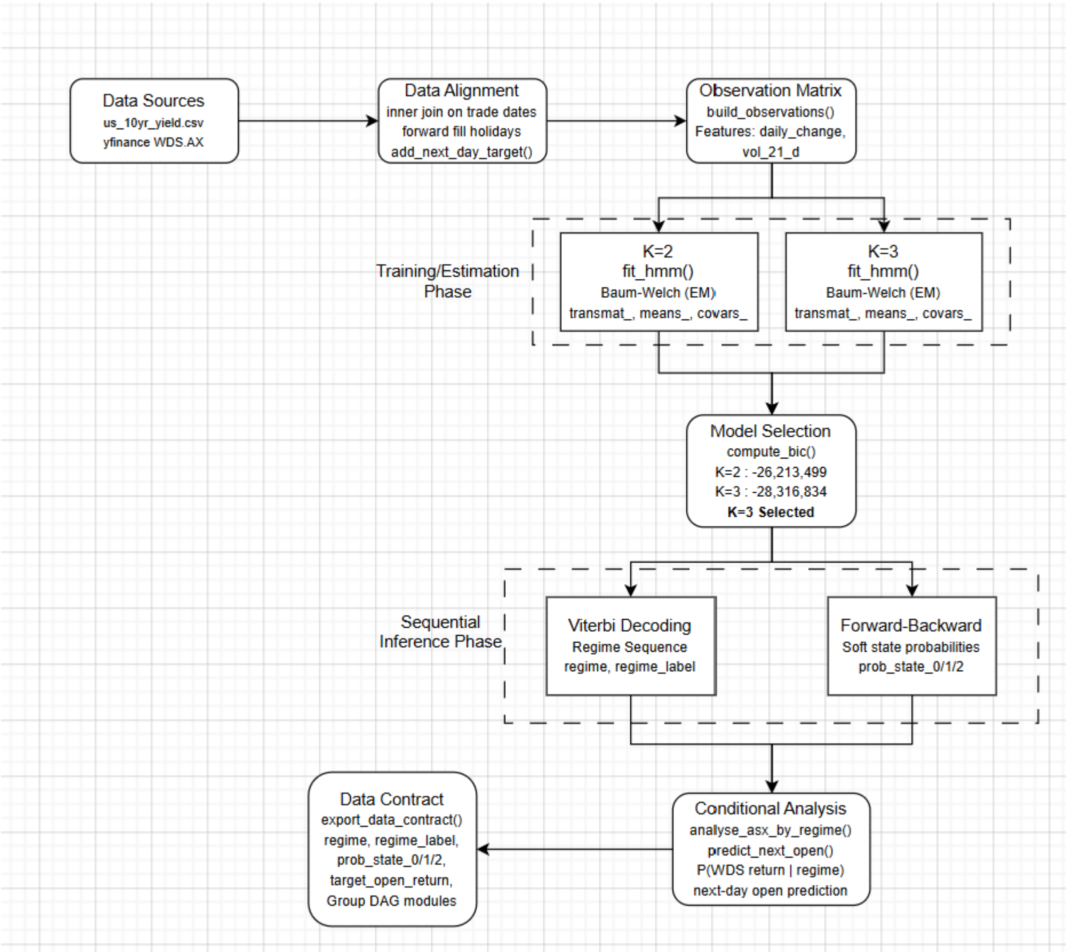
A single student-t MLE or Bayesian fit generalises to capture any given day throughout the 7-year window. There are evidently many different changes throughout the 7-year span, and a standard approach cannot capture. It also can not capture what state/regime are we in at the moment and what does this mean for uncertainty. Both of these points can be captured with a Hidden Markov Model, via Viterbi and Forward-Backward.

System Architecture and Implementation

Design



Overarching architecture combining Artefact1 and Artefact2.



Architecture for Artefact2, showing the Hidden Markov Model process.

Explain

The two data sources are loaded in, combined via inner join, forward filled data and the Woodside data is shifted back by one day to align properly. After this, 1705 trading days remain in the dataframe. The observations matrix is then built, using daily_change and vol_21d as the multivariate emission sequence. The next step is the training and estimation phase, initially K=3 was chosen due to the regimes identified in artefact1, this was put to the test using Bayesian Information Criterion (BIC) for K=2 and K=3 whilst fitting the Gaussian HMM. Baum-Welch was used for training, an unsupervised parameter estimation when state labels are unobserved. K=3 had the lower BIC score and became the chosen model. The next phase is the Sequential Inference Phase, using Viterbi decoding to find regimes and their labels and Forward-Backward process, finding the state probabilities, which give a full distributional uncertainty per day. The next step is the conditional analysis, calculating the probability of a positive return on Woodside energy stock open price per regime, and working out the P(Next-Day WDS Open Higher | Yield Regime). This analysis also returns the 95% confidence interval.

Computational Complexity and Scalability

For the model parameters T=1705(trading days), K=3(hidden states);

Baum-Welch: During training, each iteration is O(T·K^2) in time and O(T·K) in space, as our mode converged in 14 iterations, the training cost is approximately 14xO(T·K^2). Whilst our model has only 3 hidden states, with this time complexity it means that K can grow exponentially and have a massive cost as more hidden states are introduced.

Due to the smaller nature of the hidden states and time cost, this is why an exact measure was chosen rather than an approximation. The entire process runs in under a minute, a very feasible response time. Moving forward, if changes were to be made to the model to increase the number of states, an approximation method like Variational Inference could be used. By using Variational Inference, the exact posteriors are not computed but rather an approximation of them over a distribution

Evaluation and Analysis

Model Evaluation and Analysis

BIC Model Selection: K=2 approx., -15270, K=3 approx. -16429. Lower score is considerably better; 3 states being identified is consistent with Artefact1's findings (3 regimes).

Convergence: Baum-Welch converged after 14 iterations, rapid and stable convergence. This is evidence that the initialisation avoided the local optima.

Regime Distribution: stable 602 days, transitional 586 days, hiking/crisis 517 days, this is a well-balanced distribution. No evidence of wasteless states, nor is their evidence of one state that catches all days.

Test Cases

To look at the results of our model, 5 days were chosen, 2020-03-16 (COVID Crash), 2019-06-14 (calm baseline day), 2023-10-19 (hiking peak), 2022-03-01 (large loss in daily change -> large gain on next day open) and 2024-07-11 (most uncertain day).

=== Test Case Stress Results ===

date	daily_change	vol_21d	decoded_regime	prob_state_0	prob_state_1	prob_state_2	max_uncertainty	next_day_open_return_actual
2020-03-16	-0.223	0.110048	hiking/crisis	0.00000	0.000000	1.000000	0.000000	0.779502
2019-06-14	0.002	0.040027	stable	0.99979	0.000210	0.000000	0.000210	0.428230
2023-10-19	0.084	0.081267	hiking/crisis	0.00000	0.000000	1.000000	0.000000	0.492335
2022-03-01	-0.132	0.065491	hiking/crisis	0.00000	0.011469	0.988531	0.011469	3.486729
2024-07-11	-0.087	0.062745	transitional	0.00000	0.504853	0.495147	0.495147	1.248272

By looking at this table, it can be observed that there is very little to no correlation between what happens with the US 10-Year Treasury Yield. By looking at test cases 2 and 3, there can be seen a very insignificant change in the daily mean, extremely high probabilities that the regime stays in the same state and almost identical next day open returns, however they were identified as being in two different regimes. Another case that demonstrates very little to no correlation is the difference between test case 1 and 4, in the first test case the largest decrease in daily change is observed and this results in a positive return of 0.77% increase in the Woodside Stock open price, in test case 4, the decrease in daily change of US 10-Year Treasury yield is half of that in test case 1 yet the returns are 3.4%, almost 5 times as great as test case 1.

Conclusion

Making decisions using this model is not-advised, the Forward-Backward probabilities collapse near 0 and 1 for the majority of days. The most uncertain day sits at 49% uncertainty, a coin flip. This suggests that the model expresses uncertainty bimodally. Any decision maker using this model would never see a meaningful cautious probability. Even with this probability the changes are so minute the difference is insignificant.

Regime explains volatility exceptionally well but fails when it comes to next day open predictions, the regime almost explains nothing about what will happen for the following days open price. The model is reliable in telling characterising the regime but inefficient as a signal for trading.

Emission independence violation. Covariance estimates are calculated using vol_21d and daily_change, as daily change is used to calculate the rolling 21-day volatility, this violates the conditional independence assumption.

The model did not use any data for validation, due to the rolling 21-day volatility data used, randomising the days and changing the dataset around to randomly split into a train test val split is not possible. The data needs to be relative to the days prior to it.

A key finding of this study is that regime finding is real and discoverable, however this has relatively little prediction power over the next-day directional prediction.

Future potential improvements include modelling just the daily change without the rolling volatility, although with this change there is a loss to clear regime separation quality. Another improvement would be to incorporate other Woodside energy stock prices to help influence regime discovery, by allowing this there may be more reliable next day predictive power in the model.

Appendix

Citations

[1] B. Mandelbrot, "The variation of certain speculative prices," *The Journal of Business*, vol. 36, no. 4, pp. 394–419, 1963.

[2] J. D. Hamilton, "A new approach to the economic analysis of nonstationary time series and the business cycle," Econometrica, vol 57, no.2, pp. 357-384, 1989

[3] A. Ang and G. Bekaert, "Regime switches in interest rates," Journal of Business and Economic Statistics, vol 20, no 2, pp 163-182, 2002.

[3] R. Hassan and B. Nath, "Stock Market Forecasting Using Hidden Markov Model: A New Approach," in 5th International Conference on Intelligent Systems Design and Applications (ISDA'05), Melbourne, 2005.

AI Declaration

Claude (Anthropic, claude.ai, accessed May-June 2026) was used for idea generation and structural brainstorming only. All code, written analysis, and documentation was authored independently.

AI Model Used

Tool	Model	Usage Period
Claude.ai	Claude Sonnet 4.6 (Anthropic)	May 2026

#	Prompt Summary	AI Output Summary	How I Used It
1	*Asked how to get started with Artefact 2 HMM implementation*	*AI recommended hmmlearn, explained state space design, Viterbi vs Forward-Backward vs Particle Filtering, and suggested a step-by-step implementation order*	*Used as a structural guide for planning the artefact2.py pipeline*
2	*Asked for assistance with academic citations for sequential/HMM justification*	*AI verified Hamilton (1989), Ang & Bekaert (2002), Mandelbrot (1963)*	*Used as citation references; all sources independently verified before inclusion*
3	*Asked for help structuring the report*	*AI mapped all content against the assignment rubric and generated a skeleton Word document with section-by-section bullet notes*	*Used as a structural guide; all written prose authored independently*
4	*Asked for explanation of technical concepts (BIC, HMM emissions, transition matrix.)*	*AI explained each concept in plain language connected to the specific model results*	*Used to deepen understanding before writing the relevant report sections*
5	*Asked for help developing the README file*	*AI drafted bullet point requirements and what markdown language to use*	*Used to develop the readme file*